

Event Operations Command Platform

Complete Implementation Master Plan

Definitive blueprint before development · LIVE production

Version 1.2 · 6 June 2026 · Phase 16 complete · Equipment rental removed from scope

Lead: Solutions Architect · PHP · UX · Product · Security · DB · DevOps



Implementation Master Plan — Olasentra

Status: v1.2 · Active: Gate G monitoring · Next: P15 Security → P1 Design System · Constraint: Zero-downtime phased deploys

PLATFORM FOCUS: Event Operations Command Platform — event staffing, workforce ops, attendance, compliance, messaging, payroll, invoicing, client portal, operations intelligence. **Not** equipment rental. See PLATFORM-FOCUS.pdf.

LIVE PRODUCTION RULE: Every phase ships independently. New routes wrap existing pages. Feature flags in system_settings. Full backup before each deploy. Rollback = revert FTP + restore DB snapshot.

Approved development priority (post Gate G)

1. P15 Security Hardening
2. P1 Design System
3. P2 Operations Command Center
4. P3 Event Hub
5. P5 Unified Inbox
6. P6 Auto Approval Engine
7. P7 Staff Intelligence
8. P9 Client Portal
9. P10 System Health Center (+ P18–P21)
10. P12 AI Operations Assistant

Executive Summary

Dimension	Current state	Target state	Phases
Admin pages	109 PHP files, 22 nav items	~45 screens + 14 nav + 3 hubs	P0–P5, P10–P11
Staff PWA	7-tile hub	3-tab shell (Home/Inbox/Account)	P13
Intelligence	Manual approval only	Auto-approval, trust scores, readiness	P6–P8
Client access	None	Client portal subdomain	P9
Security	6 open audit items	All HIGH closed	P15
Database	20 tables, partial migration runner	28+ tables, consolidated runner	P6–P8, P16
Total effort	—	~18–24 dev-weeks (1 senior)	P0–P17

Phase 0 — Full Implementation Analysis

Objective

Map UX Master Plan v2.0 to existing codebase. Establish reuse vs build decisions.

Business value

Eliminates duplicate work; de-risks production changes; defines exact file touch list.

0.1 Existing modules

Module	Primary files	Repository	Tables
Auth & roles	includes/auth.php, admin-capabilities.php	admin-users-repository.php	admin_users
Staff / registrations	admin/staff.php, staff-directory.php, view-staff.php	staff-repository.php	staff, staff_registrations
Events & venues	admin/events.php, event-form.php, venues.php	events-repository.php, venues-repository.php	events, venues, work_types
Attendance	admin/attendance.php, scan-checkin.php	attendance-repository.php	attendance
Work hours	admin/work-hours.php	work-hours-repository.php	attendance (hours cols)
Messaging	admin/staff-inbox.php, staff-messages.php	staff-messages.php	staff_messages
Notifications	admin/notifications.php, staff-notifications.php	notification-center.php	app_notifications
Finance	admin/invoices.php, job-records.php	commission-invoice-repository.php, job-record-repository.php	commission_invoices, saved_job_records
CMS / public	admin/website-*.php, home.php	website-content.php	system_settings
Ops / health	admin/ops-checklist.php, go-live.php	ops-checklist.php, go-live.php	system_settings
Apply satellite	apply/admin/admin/*	main-admin-bridge.php	Apply DB staff_master

Integrations	google-sheets-sync.php, google-drive-oauth.php	—	system_settings, events
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0.2 Existing APIs (12 endpoints)

API	Auth	Reuse for
api/admin-staff-search.php	Admin + staff cap	P5 inbox, P12 AI search base
api/attendance-live.php	Admin + attendance	P2 Command Center, P4 Live Ops
api/notifications.php	Admin/staff session/token	P5, P11 smart notifications
api/notifications-mark-read.php	Same	P5 inbox
api/registant-lookup.php	Public (OPEN RISK)	P15 rate limit
api/events.php, registration-options.php	Public	Unchanged
api/push-subscribe.php, push-vapid-public.php	Token	P11, P13 PWA
api/roster-check.php, health.php	Public	P10 health center

0.3 UX Master Plan → File mapping

UX feature	Existing	Modify	New	DB
Command Center	dashboard.php, getDashboardStats()	dashboard.php, layout-top.php	includes/command-center.php, api/command-center.php	Indexes only
Activity & Inbox	notifications.php, staff-inbox.php	notification-center.php, staff-messages.php	inbox.php, assets/js/inbox.js	app_notifications.type enum extend
Event Hub	events.php, staff.php, attendance.php...	Add event_id filter to all	event.php, includes/event-hub.php	None
Live Ops	scan-checkin.php, attendance-live API	scan-checkin-action.php	live-ops.php, assets/js/live-ops.js	attendance.incident_note (optional)
Command palette	header-bar.php erp-search	header-bar.php, admin.js	api/admin-search.php, assets/js/command-palette.js	None
Design system OOU	admin-sidebar.css, admin.css	All admin CSS	assets/css/ops-ui.css, includes/components/ops-*.php	None
Auto approval	update-status.php, staff-psa.php, blacklist	submit.php, staff-repository.php	includes/auto-approval-engine.php	approval_rules, approval_log tables
Staff trust score	attendance, blacklist partial	staff-directory.php	includes/staff-intelligence.php	staff_trust_scores table
Event readiness	ops-checklist.php, go-live.php	ops-checklist.php	includes/event-readiness.php	event_readiness_snapshots
Client portal	None	—	client/* entire subtree	client_users, client_event_access
System health	sidebar-health.php, go-live.php	go-live.php	admin/system-health.php	system_health_checks log
AI assistant	admin-staff-search partial	—	api/ai-ops.php, includes/ai-ops.php	ai_query_log
Staff PWA 2.0	staff-app.php, staff-messages.php	staff-app.php, pwa-head.php	staff-home.php, staff-account.php	None
Public redesign	home.php, website-*.php	All public pages	assets/css/public-premium.css	system_settings CMS keys
Security hardening	auth.php, .htaccess	apply/admin/*, api/registant-lookup.php	includes/rate-limiter.php, apply CSRF helper	rate_limit_log

0.4 Key dependencies

- **Composer:** web-token/jwt-library (push), guzzlehttp (HTTP), spomky-labs/pki-framework — do not break vendor/
- **External:** Google Sheets/Drive OAuth, SMTP, VAPID push, Apply SSO, FTP deploy
- **Runtime schema:** 24 ensure*Schema() functions — consolidate in P16 before adding more
- **Subdomains:** admin.*, register.*, apply.* — client.* new in P9

Phase 0 deliverables

Effort: **Complete (this document)** · Risk: None · Rollback: N/A

Phase 1 — Design System Foundation (OOU)

Objective

Establish Olasentra Ops UI tokens, primitives, and component PHP partials used by all subsequent phases.

Business value

Consistent high-density ops UI; faster phase delivery; one source of truth for Linear/Stripe aesthetic.

Design tokens (CSS custom properties)

```
assets/css/ops-ui.css (NEW — loaded after admin.css)
--ops-bg-base #030712      --ops-bg-canvas #f8fafc / #0b0f19
--ops-accent #8b5cf6        --ops-live #22c55e      --ops-warn #f59e0b      --ops-critical #ef4444
--ops-text-xs 11px          --ops-row-h 32px        --ops-radius 6px
Font: Inter (Google) + JetBrains Mono for KPIs
```

Files to update

File	Change
assets/css/admin.css	Import ops-ui.css; deprecate duplicate vars
assets/css/admin-sidebar.css	Align tokens to ops-ui.css
includes/admin/layout-top.php	Load ops-ui.css, Inter font
assets/css/messages.css, notifications.css	Extend with OOU badge/thread styles
assets/js/admin.js	Extract palette prep; move sidebar to admin-sidebar.js (done)

New files

- assets/css/ops-ui.css — tokens + utilities
- includes/components/ops-button.php — renderOpsButton()
- includes/components/ops-badge.php — status pills
- includes/components/ops-data-table.php — dense table wrapper
- includes/components/ops-kpi.php — ribbon + card KPIs
- includes/components/ops-attention-card.php — severity stack cards
- includes/components/ops-drawer.php — right panel shell
- includes/components/ops-empty-state.php
- assets/js/ops-ui.js — drawer, tabs, toast helpers

Component library scope (22 components — see Phase 1 checklist in UX Master Plan)

Modal: extend existing admin flash pattern. Form: reuse settings form classes. Notification: extend notification-list.php.

Database / APIs

None.

Risks

CSS regression on 109 pages. Mitigation: scoped .ops-* prefix; visual regression on dashboard + staff + events.

Rollback

Remove ops-ui.css from layout-top.php. Revert CSS files from backup.

Testing

- Dashboard, staff list, events, scan-checkin render correctly light+dark
- Sidebar collapse 280px→72px unchanged
- Mobile admin drawer works
- Staff PWA unaffected (separate CSS)

Deploy

CSS-only deploy. No DB. Clear browser cache note in ops checklist.

Effort: 1 week · Priority: **P0 blocker for P2–P5**

Phase 2 — Command Center

Objective

Replace dashboard.php hero layout with SOC-style Operations Command Center.

Business value

Ops managers see prioritized actions in <3 seconds; event-day CTA auto-surfaces; reduces navigation by 40%.

Features → implementation

Feature	Existing code	Implementation
Status Ribbon	getDashboardStats(), getTodayCheckinCount()	includes/command-center.php → getCommandCenterRibbon()
Attention Stack	dashboard.php pending preview, countUnread*	getCommandCenterAttentionStack() — max 5 severity-sorted
Event Timeline	getUpcomingEventsSummary()	Extend with approved/pending/on-site per event
Live Activity Feed	api/attendance-live.php	Embed poll 30s on event day; reuse getLiveAttendancePayload()
Event Context Selector	staff.php event filter	Header OpsEventSelector; persist ?event_id= + localStorage
Quick Actions	dashboard quick strip	Role-filtered; single primary CTA from attention stack
Dashboard Intelligence	ops-checklist getOpsLiveMetrics()	Merge ops metrics into attention stack

Files affected

Modify: admin/dashboard.php (full restructure), includes/admin/header-bar.php (event selector), includes/staff-repository.php (new queries), includes/admin-capabilities.php (nav label "Command Center")

New: includes/command-center.php, api/command-center.php (JSON poll for ribbon/feed), assets/js/command-center.js

Database

Recommended index: `staff_registrations(event_id, status)` — migration phase 48

APIs

- `GET api/command-center.php?event_id=` — ribbon, attention, timeline, feed (admin session)

Risks

Performance on large registration counts. Mitigation: indexed queries, limit feed to 20 rows, cache timeline 5min.

Rollback

Revert dashboard.php from backup. Remove api/command-center.php.

Testing

- Admin/Manager/Staff role see correct panels
- Event selector filters all panels
- Attention stack links work (pending→staff.php, messages→inbox.php)
- Live feed activates only on event day window
- Nick Cave production: 36 registrations display correctly

Effort: 1.5 weeks · Depends: P1

Phase 3 — Event Detail Hub (event.php)

Objective

Single event command page with 8 tabs embedding existing functionality.

Business value

One URL per event for all ops; eliminates 8 disconnected nav hops.

Tab mapping

Tab	Embeds existing	Query param
Overview	NEW stats panel	event.php?id=X
Roster	admin/staff.php	?event_id=X&embed=1
Attendance	admin/attendance.php	?event_id=X&embed=1
Hours	admin/work-hours.php	?event_id=X&embed=1
QR & Sign-in	print-qr.php, event-sign-qr.php	inline panels
Sheets	events-sheets-action UI	from events.php row actions
Finance	invoice-quick.php, job-records filtered	event_id filter
Config	event-form.php fields read-only + quick edit	work_types, roles_needed

Files

New: admin/event.php, includes/event-hub.php, assets/js/event-hub.js, assets/css/event-hub.css

Modify: admin/events.php (link→event.php), staff.php/attendance.php/work-hours.php (embed mode: hide sidebar via `$embedMode`), includes/admin/layout-top.php (embed layout variant)

Database

None required. Optional: `events.briefing_sent_at` for readiness (P8).

APIs

`GET api/event-overview.php?id=` — stats JSON for Overview tab

Risks

Embed mode CSS conflicts. Mitigation: `.ops-embed-layout` body class hides sidebar/header.

Rollback

events.php links back to old URLs. Delete event.php.

Testing

- All 8 tabs load with correct event filter
- Approve from Roster tab works
- Print QR from QR tab
- Deep link event.php?id=1 matches Nick Cave data

Effort: 1.5 weeks · Depends: P1, P2 optional

Phase 4 — Live Ops Mode (live-ops.php)

Objective

Full-screen event-day operations: scanner + roster board + live ticker.

Business value

Venue staff check in 3× faster; real-time on-site visibility; flight-control UX.

Features

Feature	Source
QR Scanner	admin/scan-checkin.php + scan-checkin-action.php

Live Check-ins	api/attendance-live.php (poll 10s)
Attendance Board	attendance-repository getAttendanceList filtered
Roster Board	staff.php approved list with on-site status
Incident Tracking	NEW optional attendance.incident_note column
Real-Time Updates	Extend attendance-live.php with WebSocket future; poll MVP

Files

New: admin/live-ops.php, assets/js/live-ops.js, assets/css/live-ops.css, includes/live-ops.php

Modify: admin/scan-checkin.php (redirect to live-ops or embed), includes/admin-capabilities.php (nav: Live Ops), scan-checkin-action.php (return JSON for AJAX scanner)

Database

Optional: ALTER TABLE attendance ADD incident_note VARCHAR(255) NULL — phase 49

APIs

- Extend api/attendance-live.php — add roster board payload (approved + checked_in flag)
- POST admin/scan-checkin-action.php — ensure JSON response mode

Auto-activation

includes/live-ops.php → shouldActivateEventDayMode(\$pdo) — checks event window; Command Center CTA links here.

Risks

Camera permissions on mobile admin PWA. Scanner failure fallback to manual check-in drawer.

Rollback

scan-checkin.php remains standalone. Remove live-ops nav item.

Testing

- QR scan records attendance (existing flow)
- Roster board updates within 10s
- Manual check-in drawer works
- Staff role lands on Live Ops by default
- Progress bar matches on-site count

Effort: 2 weeks · Depends: P1, P3 recommended

Phase 5 — Unified Inbox (inbox.php)

Objective

Merge notifications.php + staff-inbox.php into Activity & Inbox with master-detail layout.

Business value

Single badge; one habit for ops comms; 50% fewer missed alerts.

Views

View	Data source	Existing functions
All	Merged timeline	NEW mergeInboxTimeline()
Messages	staff_messages	listStaffInboxThreads(), getStaffMessageThread()
Activity	app_notifications admin	getAdminNotifications()
System	app_notifications type=system	Filter by type

Files

New: admin/inbox.php, includes/inbox-repository.php, assets/js/inbox.js

Modify: includes/admin-capabilities.php (replace notifications+staff-inbox nav with inbox), includes/admin/sidebar.php (single badge), admin/notifications.php + staff-inbox.php (301→inbox.php?view=), includes/notification-center.php (add category field support P11)

Database


```
ALTER TABLE app_notifications ADD category ENUM('staff','events','attendance','finance','security','system') NULL;  
ALTER TABLE app_notifications ADD priority TINYINT UNSIGNED DEFAULT 2;
```

APIs

- GET `api/inbox.php?view=&event_id=&q=`
- POST `api/inbox-mark-read.php`

Risks

Timeline merge performance. Mitigation: paginate 50, index (audience, is_read, created_at) exists.

Rollback

Restore sidebar links to notifications.php + staff-inbox.php.

Testing

- Unread badge = messages + activity combined
- Reply in thread without page reload
- Activity CTA opens correct target
- Filters: unread, today, by event

Effort: **1.5 weeks** · Depends: P1, P11 partial overlap

Phase 6 — Auto Approval Engine

Objective

Intelligent registration routing: auto-approve, manager review, incomplete, auto-reject.

Business value

Reduce manager workload 60% for repeat staff; faster roster fill; compliance gate automated.

Check pipeline (ordered)

1. Blacklist → Auto Rejected
2. Duplicate email+event → Auto Rejected
3. Profile completion (staff.profile_completed) → Incomplete (hold)
4. PSA validity (staff.psa_expiry_date) → Incomplete or Manager Review
5. Event roles_needed match → Manager Review if mismatch
6. Staff trust score < threshold (P7) → Manager Review
7. All pass + trust ≥ Gold/Silver → Auto Approved

Files

New: includes/auto-approval-engine.php, includes/approval-rules.php, admin/approval-rules.php (settings UI)

Modify: submit.php (run engine post-insert), admin/update-status.php (log override), includes/staff-repository.php (bulk path), includes/notification-center.php (notify outcome)

Database

```
CREATE TABLE approval_rules (
  id INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  rule_key VARCHAR(50) NOT NULL UNIQUE,
  is_enabled TINYINT(1) DEFAULT 1,
  config_json JSON NULL,
  updated_at TIMESTAMP
);
CREATE TABLE approval_log (
  id INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  registration_id INT UNSIGNED NOT NULL,
  outcome ENUM('auto_approved', 'manager_review', 'incomplete', 'auto_rejected') NOT NULL,
  rule_results JSON NOT NULL,
  decided_by ENUM('system', 'admin') DEFAULT 'system',
  admin_user_id INT UNSIGNED NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  INDEX idx_approval_reg (registration_id)
);
```

APIs

POST api/approval-preview.php — dry-run for admin on view-staff.php

Risks

False auto-approve on production. Mitigation: feature flag `auto_approval_enabled=0` **default OFF; shadow mode logs only for 2 weeks.**

Rollback

Disable flag. Engine skipped in submit.php. approval_log retained for audit.

Testing

- Blacklisted email → rejected
- Incomplete profile → pending + profile_required notification
- Valid returning staff → auto-approved (flag on, staging only first)
- Manager override logged in approval_log

Effort: **2 weeks** · Depends: P7 trust score optional MVP without

Phase 7 — Staff Intelligence

Objective

Staff Trust Score (Gold/Silver/Standard/Review) + event staffing recommendations.

Business value

Data-driven roster decisions; reduce no-shows; premium staff prioritized.

Trust score model

Tier	Score range	Criteria weights
Gold	85–100	Attendance 40%, compliance 30%, tenure 20%, ratings 10%
Silver	70–84	Same formula
Standard	50–69	Default new staff
Review Required	<50 or flag	No-shows, blacklist history, PSA gaps

Metrics sources

- Attendance rate: attendance vs approved registrations per staff_id
- No-shows: approved + no attendance record after event end
- Compliance: staff.psa_expiry_date, profile_completed
- Event history: count approved registrations
- Manager ratings: NEW manual 1–5 on view-staff.php

AI recommendations (rule-based MVP, not LLM)

`suggestStaffForEvent($pdo, $eventId, $role, $limit)` — ranks by trust score + distance (eircode) + role match. Display on event.php Overview tab.

Files

New: includes/staff-intelligence.php, cron/compute-trust-scores.php, admin/staff-intelligence.php (report)

Modify: admin/staff-directory.php (trust badge), admin/view-staff.php (rating input), event.php Overview (suggestions panel)

Database

```
CREATE TABLE staff_trust_scores (
  staff_id INT UNSIGNED PRIMARY KEY,
  score SMALLINT UNSIGNED NOT NULL DEFAULT 50,
  tier ENUM('gold','silver','standard','review') NOT NULL DEFAULT 'standard',
  attendance_rate DECIMAL(5,2) NULL,
  events_worked INT UNSIGNED DEFAULT 0,
  no_show_count INT UNSIGNED DEFAULT 0,
  manager_rating_avg DECIMAL(3,2) NULL,
  computed_at TIMESTAMP,
  FOREIGN KEY (staff_id) REFERENCES staff(id)
);
CREATE TABLE staff_manager_ratings (
  id INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  staff_id INT UNSIGNED NOT NULL,
  event_id INT UNSIGNED NULL,
  admin_user_id INT UNSIGNED NOT NULL,
  rating TINYINT UNSIGNED NOT NULL,
  note VARCHAR(255) NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Risks

Score perceived as unfair. Mitigation: transparent breakdown on directory; manual tier override for admin.

Rollback

Hide badges UI. Cron stops. Tables retained.

Effort: **2 weeks** · Depends: P16 indexes, P6 optional integration

Phase 8 — Event Readiness Engine

Objective

Event Readiness Score (0–100) evaluating staffing, compliance, roster, sheets, briefing, venue.

Business value

Pre-event confidence metric; ops checklist automated; client reporting ready.

Scoring model (100 points)

Factor	Max pts	Source
Staffing level	25	approved / staff_needed
Compliance	20	% approved with valid PSA
Roster completion	20	pending cleared
Sheets sync	15	google_sheet_url set + last sync ok
Briefing status	10	briefing_sent_at or manual checklist
Venue readiness	10	venue GPS + eircode + times_confirmed

Files

New: includes/event-readiness.php

Modify: includes/ops-checklist.php (integrate score), admin/event.php Overview, admin/ops-checklist.php, Command Center attention stack

Database

```
CREATE TABLE event_readiness_snapshots (  
  id INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  event_id INT UNSIGNED NOT NULL,  
  score TINYINT UNSIGNED NOT NULL,  
  factors_json JSON NOT NULL,  
  computed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  INDEX idx_readiness_event (event_id, computed_at)  
);  
ALTER TABLE events ADD briefing_sent_at TIMESTAMP NULL;
```

APIs

```
GET api/event-readiness.php?event_id=
```

Risks

Low — read-only computation.

Effort: 1 week · Depends: P3 event hub

Phase 9 — Client Portal

Objective

Separate client-facing portal for event clients to view rosters, attendance, invoices, request staff.

Business value

Revenue retention; reduces email/WhatsApp reporting; professional client experience.

Architecture

```
client.olasentra.com (NEW subdomain)
client/
  index.php, login.php, logout.php
  dashboard.php – client events list
  event.php – roster + attendance read-only
  invoices.php – commission invoices for their events
  reports.php – PDF/CSV download
  request-staff.php – staffing request form → admin notification
  messages.php – thread with ops (optional P5 integration)
```

Permissions model

Role	Access
client_admin	All events for client_org
client_viewer	Read-only assigned events

Database

```
CREATE TABLE client_organisations (
  id INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(150) NOT NULL,
  contact_email VARCHAR(150) NOT NULL,
  is_active TINYINT(1) DEFAULT 1
);
CREATE TABLE client_users (
  id INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  org_id INT UNSIGNED NOT NULL,
  email VARCHAR(150) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  role ENUM('admin','viewer') DEFAULT 'viewer',
  FOREIGN KEY (org_id) REFERENCES client_organisations(id)
);
CREATE TABLE client_event_access (
  client_org_id INT UNSIGNED NOT NULL,
  event_id INT UNSIGNED NOT NULL,
  PRIMARY KEY (client_org_id, event_id)
);
CREATE TABLE client_staff_requests (
  id INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  org_id INT UNSIGNED NOT NULL,
  event_id INT UNSIGNED NULL,
  body TEXT NOT NULL,
  status ENUM('open','acknowledged','closed') DEFAULT 'open',
  created_at TIMESTAMP
);
```

Files (all NEW)

~15 PHP files under client/, includes/client-auth.php, includes/client-repository.php, assets/css/client-portal.css

Modify: admin/event-form.php (assign client_org), includes/commission-invoice-repository.php (client filter)

Risks

PII exposure to clients (roster names). Mitigation: configurable field visibility per org; GDPR consent.

Rollback

Subdomain not configured = portal unreachable. No impact on main ops.

Effort: 3 weeks · Depends: P3, P15 security patterns

Phase 10 — System Health Center

Objective

Consolidate go-live.php + sidebar-health into full infrastructure monitoring dashboard.

Monitors

Check	Existing	Implementation
Database	api/health.php, sidebar-health	PDO ping + table count
Storage	api/health.php	writable storage/ check
Email	settings test send	SMTP connect test
Google Sheets	google-sheets-diagnostic.php	Token + sample read
Google Drive	google-drive-oauth.php	OAuth expiry check
Push	pwa-push.php VAPID	Key presence
SSL	NEW	curl https self-check
Backups	admin/backups.php	Last backup age
Cron	ops-checklist	daily-cleanup last run log
Apply Sync	apply-sync-ping.php	SSO + staff count compare

Files

New: admin/system-health.php, includes/system-health.php, api/system-health.php

Modify: admin/go-live.php (link→system-health), includes/admin/sidebar-health.php (delegate), includes/ops-checklist.php

Database

```
CREATE TABLE system_health_log (  
  id INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  check_key VARCHAR(50) NOT NULL,  
  status ENUM('ok','warn','fail') NOT NULL,  
  message VARCHAR(255) NULL,  
  checked_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  INDEX idx_health_time (checked_at)  
);
```

Alerting

Failed check → app_notifications (audience=admin, category=system) + optional email if 2 consecutive fails.

Risks

Google API quota on frequent checks. Mitigation: cache 15min per check.

Effort: 1 week · Depends: P1, P5 notifications

Phase 11 — Smart Notifications

Objective

Upgrade notification system with categories, priority, avatars, event badges, bulk actions, real-time.

Files

Modify: includes/notification-center.php (category/priority on create), includes/components/notification-list.php (avatar, badges), api/notifications.php (filter params), admin/inbox.php Activity view, staff-notifications.php

New: assets/js/notifications-realtime.js (poll 60s admin, 120s staff)

Database

Phase 5 category/priority columns. Add: `app_notifications.event_id INT UNSIGNED NULL`

Backfill categories

type prefix	category
registration*, status*	staff
event_*	events

checkin*	attendance
invoice*	finance

Risks

Low — extends existing table.

Effort: 1 week · Depends: P5 inbox

Phase 12 — AI Operations Assistant

Objective

Design Olasentra AI — ops query interface. MVP: structured queries, not generative LLM on PII.

MVP capabilities (rule-based + SQL templates)

Query type	Data source
"How many approved for Nick Cave?"	staff_registrations count
"Staff with expiring PSA"	staff.psa_expiry_date
"Event readiness"	event-readiness.php
"Payroll summary June"	attendance hours aggregate
"Suggest stewards for event X"	staff-intelligence.php

Architecture (future LLM-ready)

```
admin AI drawer (🔑 → "Ask Olasentra")
  → api/ai-ops.php (admin session, rate limited)
  → includes/ai-ops.php query router
    → approved SQL templates ONLY (no raw LLM SQL)
    → log to ai_query_log
Optional Phase 12b: OpenAI API with anonymized aggregates only
```

Security model

- Admin role only (not manager initially)
- No PPS/IBAN in AI responses — mask sensitive fields
- 10 queries/min rate limit
- Full audit log
- PII fields excluded from any future LLM context

Files

New: includes/ai-ops.php, api/ai-ops.php, assets/js/ai-ops-drawer.js, admin/ai-ops-settings.php

Database

```
CREATE TABLE ai_query_log (
  id INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  admin_user_id INT UNSIGNED NOT NULL,
  query_text VARCHAR(500) NOT NULL,
  intent VARCHAR(50) NULL,
  response_summary VARCHAR(255) NULL,
  created_at TIMESTAMP
);
```

Risks

Scope creep into LLM. Mitigation: MVP ships template queries only; LLM is Phase 12b gated on privacy review.

Effort: 2 weeks MVP · Depends: P5, P7, P8

Phase 13 — Staff PWA 2.0

Objective

3-tab shell: Home / Inbox / Account for signed-in staff.

Screen map

Tab	Route	Source
Home	staff-home.php (NEW)	Next shift from staff_registrations; status chips; broadcast
Inbox	staff-inbox.php (NEW) or staff-messages.php evolve	Merge staff-messages + staff-notifications
Account	staff-account.php (NEW)	staff-profile.php compact + PSA + sign out

Files

New: staff-home.php, staff-inbox.php, staff-account.php, includes/staff-app-nav.php, assets/css/staff-tabs.css, assets/js/staff-app.js

Modify: staff-app.php (redirect signed-in→staff-home), includes/staff-portal-session.php (landing URL), staff-messages.php + staff-notifications.php (301→staff-inbox), sw.js (cache shell), includes/pwa-head.php

Database

None.

Risks

Token-based access for unsigned users must remain on status.php / inbox ?token=.

Testing

- Install PWA on iOS/Android
- Push notification on approval
- Next shift card accurate
- Sign out returns to staff-app.php hub

Effort: **2 weeks** · Depends: P5 inbox patterns, P11 notifications

Phase 14 — Public Website Redesign

Objective

Premium marketing site — Apple/Netflix/Eventbrite aesthetic.

Pages

Page	File	Change
Home	home.php	Cinematic hero, event carousel, social proof
Events	events-page.php	Card grid from live events table
Join Team	roles.php + CTA→index.php	Role explainer video/embed
Services	NEW services.php	B2B client offering (feeds P9)
About	NEW about.php or CMS section	Company story
Contact	contact.php	Premium form

Files

New: assets/css/public-premium.css, assets/js/public-premium.js, services.php, about.php

Modify: includes/public/layout-top.php, all public pages, admin/website-*.php CMS keys

Database

system_settings: new CMS keys for services/about hero content.

Risks

SEO disruption. Mitigation: preserve URLs; 301 only for new pages.

Effort: **2 weeks** · Independent of admin ops phases

Phase 15 — Security Hardening

Objective

Close all HIGH audit items before intelligence/client portals expand attack surface.

Task list → files

Task	Files
Apply admin CSRF	apply/admin/includes/csrf.php (NEW), all apply/admin/admin/*.php POST forms, update-details.php
Upload validation	apply/admin/update-details.php — MIME, size, rename, CSRF
Rate limiting	includes/rate-limiter.php (NEW), api/registrant-lookup.php, api/ai-ops.php
Session hardening	apply/admin/includes/auth.php → initSecureSession()
SSO secrets	apply/admin/config/sso.local.php (deploy), remove dev fallback in apply-sso.php
GET→POST	admin/import-roster.php, apply/admin/admin/sync-sheets.php, admin/backups-export.php
Apply uploads htaccess	apply/admin/uploads/.htaccess (NEW)
API protection	api/*.php — X-Requested-With check for JSON endpoints

Database

```
CREATE TABLE rate_limit_log (  
  ip_hash CHAR(64) NOT NULL,  
  endpoint VARCHAR(80) NOT NULL,  
  hit_count INT UNSIGNED DEFAULT 1,  
  window_start TIMESTAMP,  
  PRIMARY KEY (ip_hash, endpoint)  
);
```

Risks

Apply CSRF breaks bookmarked POST flows. Mitigation: staging test on apply.olasentra.com first.

Rollback

Revert apply/admin files. Rate limiter disable flag in settings.

Effort: 1.5 weeks · Priority: **Before P9 client portal**

Phase 16 — Database Optimization (PACKAGE READY)

Full package: docs/PHASE-16-IMPLEMENTATION-PACKAGE.pdf · SQL artifacts in database/

Objective

Consolidate migration runner (phases 11–13, 30–31, 35–51), composite indexes, staff_id backfill, FK gaps, platform ops tables for approved future modules.

SQL files (generated — not yet applied to production)

File	Purpose
database/phase16-precheck.sql	Read-only diagnostics BEFORE migrate
database/migrate-phase48-composite-indexes.sql	Performance indexes
database/migrate-phase49-backfill-staff-id.sql	Link registrations → staff
database/migrate-phase50-foreign-keys.sql	FK staff_messages, invoice lines, job records
database/migrate-phase51-platform-ops-tables.sql	Cleanup, storage, audit, SSL, emergency (equipment tables dormant — out of scope)
database/phase16-postcheck.sql	Verification AFTER migrate
database/rollback-phase48-51.sql	Rollback indexes, FKs, new tables
docs/PHASE-16-RUNNER-MANIFEST.txt	Exact getDatabaseMigrationFiles() order

PHP changes (Phase 16b — after SQL approved)

- includes/database-migrations.php — runner manifest (52 files)
- includes/staff-repository.php — deleteStaffProfileCompletely() + messages/notifications/push
- admin/go-live.php — migration status panel

Execution order

Backup → precheck → runner gap SQL (35–47 if missing) → phase48 → phase49 → precheck orphans → phase50 → phase51 → postcheck → deploy PHP runner update.

Effort: 1 week · **Gate F:** Approve PHASE-16-IMPLEMENTATION-PACKAGE before production SQL run · Phase 15 blocked until P16 complete

Phase 18 — Cleanup Center

Objective: Unified admin surface for safe log/temp/cache cleanup (evolves system-cleanup.php).

Reuse: admin/system-cleanup.php, includes/system-cleanup.php (getStorageUsageReport, runSystemCleanup)

New: admin/cleanup-center.php, includes/cleanup-center.php · **DB:** platform_cleanup_log (phase 51)

Effort: 1 wk · Depends: P16, P10

Phase 19 — Storage Analyzer

Objective: Deep disk analysis with trends, large-file finder, backup rotation advisor.

Reuse: getStorageUsageReport(), directorySizeBytes(), admin/backups.php

New: Tab in Cleanup Center · cron/monthly-storage-snapshot.php · **DB:** platform_storage_snapshots

Effort: 0.5 wk · Depends: P18

Phase 20 — Monthly System Audit

Objective: Automated monthly audit report (security, DB, backups, registrations, integrations).

Reuse: docs/AUDIT-REPORT template, go-live.php checks, ops-checklist.php

New: cron/monthly-system-audit.php, includes/monthly-audit.php, admin/audit-reports.php · **DB:** platform_audit_runs

Effort: 1 wk · Depends: P16, P10

Phase 21 — Domain & SSL Monitoring

Objective: Monitor admin.*, register.*, apply.*, main domain — SSL expiry, HTTP reachability.

Reuse: includes/site-urls.php, api/health.php pattern

New: includes/ssl-monitor.php, cron/ssl-check.php, System Health panel · **DB:** platform_ssl_checks

Effort: 0.5 wk · Depends: P10, P16

Phase 22 — Emergency Event Mode

Objective: One-click escalation: mass notify, lock roster, fullscreen Live Ops, incident log.

Reuse: live-ops.php (P4), event-staff-alerts.php, staff-broadcast.php

New: admin/emergency-event.php, includes/emergency-event.php · **DB:** emergency_event_log · Flag: feature_emergency_event_mode

Effort: 1 wk · Depends: P4, P11

Removed from roadmap (v1.2): Equipment Rental (former P23). Tables `equipment_categories`, `equipment_items`, `equipment_rentals` may exist empty on production — **no development** unless explicitly approved later. Olasentra is not an equipment rental platform.

Phase 17 — Deployment Strategy

Deployment pipeline (every phase)

1. Admin → Backups → Full backup (production)
2. scripts/pre-deploy-backup.ps1 (local zip)
3. Git commit (user-requested only)
4. Staging verification if available (Laragon local)
5. powershell -ExecutionPolicy Bypass -File .\deploy.ps1

6. Post-deploy smoke test (see checklist per phase)
7. Monitor system-health + error logs 24h

Phase deployment order (recommended)

Wave	Phases	Rationale
Wave 1 Foundation	P16 ✓ → Gate G → P15 → P1	DB complete; security; design system
Wave 2 Core Ops	P2, P3, P4, P5	Command Center, Event Hub, Live Ops, Inbox
Wave 3 Intelligence	P6, P7, P8	Auto approval, staff intelligence, readiness
Wave 4 Health & Ops	P10, P18–P22	System Health Center, cleanup, SSL, emergency
Wave 5 Growth	P9, P13, P12	Client portal, Staff PWA 2.0, AI ops assistant
Wave 6 Marketing	P14	Public website refresh

Feature flags (system_settings keys)

```
feature_command_center_v2=0
feature_event_hub=0
feature_live_ops=0
feature_unified_inbox=0
feature_auto_approval=0
feature_trust_scores=0
feature_client_portal=0
feature_ai_ops=0
feature_staff_pwa_v2=0
```

Global rollback procedure

1. Restore FTP from pre-deploy-backup zip (scripts/upload-to-server.ps1 reverse manual)
2. Restore MySQL from admin/backups download
3. Clear opcache if applicable
4. Verify api/health.php

Global production smoke test

- admin login + dashboard load
- staff.php Nick Cave 36 rows
- register.olasentra.com form submit (test email)
- scan-checkin.php camera init
- staff-app.php load
- apply SSO bridge

Master Effort Summary

Phase	Name	Effort	Priority	Depends
P0	Analysis	Done	—	—
P1	Design System OOU	1 wk	Critical	—
P2	Command Center	1.5 wk	Critical	P1
P3	Event Hub	1.5 wk	High	P1
P4	Live Ops	2 wk	Critical	P1,P3
P5	Unified Inbox	1.5 wk	High	P1
P6	Auto Approval	2 wk	High	P16
P7	Staff Intelligence	2 wk	Medium	P16
P8	Event Readiness	1 wk	Medium	P3
P9	Client Portal	3 wk	Medium	P15,P3
P10	System Health	1 wk	High	P1
P11	Smart Notifications	1 wk	Medium	P5
P12	AI Ops Assistant	2 wk	Medium	P5,P7,P8
P13	Staff PWA 2.0	2 wk	High	P5,P11
P14	Public Website	2 wk	Medium	—
P15	Security Hardening	1.5 wk	Critical	—
P16	Database Optimization	1 wk	Critical	—
P17	Deployment Strategy	Ongoing	—	All
P18	Cleanup Center	1 wk	High	P16,P10
P19	Storage Analyzer	0.5 wk	Medium	P18
P20	Monthly System Audit	1 wk	High	P16,P10
P21	Domain & SSL Monitoring	0.5 wk	High	P10,P16
P22	Emergency Event Mode	1 wk	High	P4,P11
Total		~27 weeks	P16 complete · P15 after Gate G · P23 rental removed	

Approval Gates

1. **Gate A:** UX Master Plan v2 + Implementation Plan v1.0 — *APPROVED*
2. **Gate F (before P16 production SQL):** Approve PHASE-16-IMPLEMENTATION-PACKAGE.pdf
3. **Gate G (before P15):** P16 postcheck passed on production + 24h stable
4. **Gate B (before Wave 2):** P1 component library reviewed on staging
5. **Gate C (before P6 auto-approve):** Shadow mode 2 weeks minimum on production
6. **Gate D (before P9 client portal):** P15 security audit re-run; GDPR review
7. **Gate E (before P12 LLM):** Privacy impact assessment; MVP templates only until approved